

Elasticity of Substitution Between Public and Private Goods

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Sufficient statistic #1: Elasticity of substitution b/w public & private goods, ε

Definition:

$$\frac{1}{\varepsilon} = - \frac{\partial \ln(MRS_{gc})}{\partial \ln(g/c)}$$

• MRS_{gc} is $\downarrow$ in g/c so $\varepsilon > 0$

• 3 different cases

• $\varepsilon < 1$: public & private goods are gross complement

• $\varepsilon = 1$: public & private goods are independent

• $\varepsilon > 1$: public & private goods are gross substitute

• 3 special cases

• $\varepsilon \rightarrow 0$: perfect complements

ex. Leontief : $U(c, g) = \min(c, g)$

• $\varepsilon = 1$: independent

ex. $U(c, g) = c^{1-\sigma} g^{\sigma}$: Cobb-Douglas

. $\varepsilon \rightarrow 0$:

perfect substitute

ex: linear . $u(c, g) = c + g$